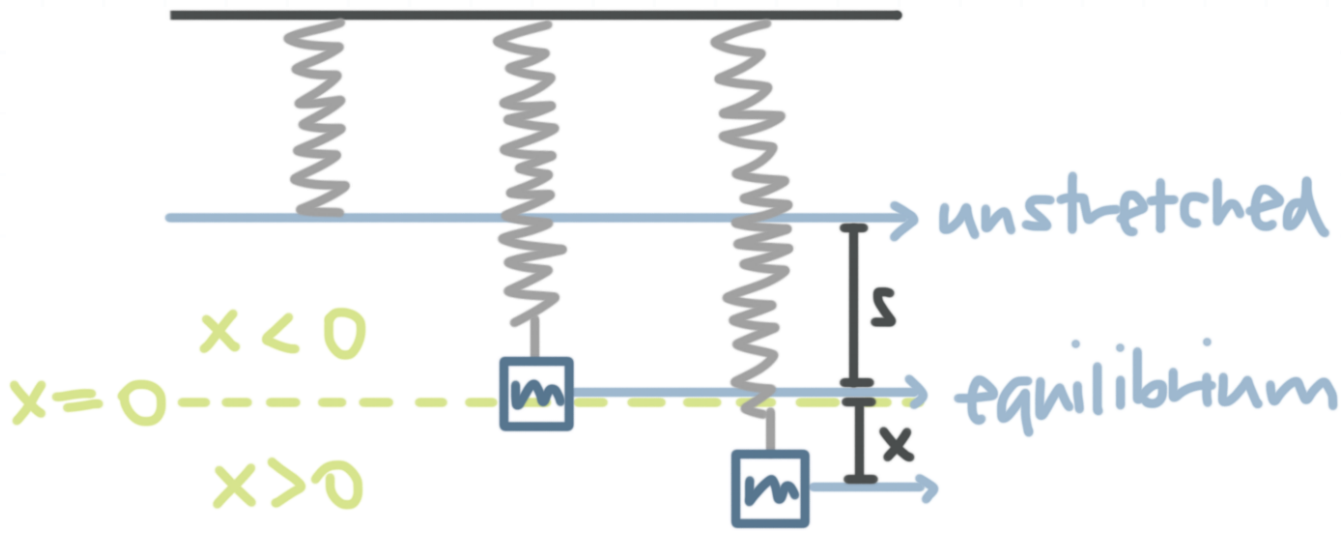
Spring and mass systems

Spring and mass systems are another common real-world application of differential equations. Like the electrical circuits we looked at earlier, these also require that we solve second order equations.

The general idea here is that we'll have some unstretched spring hanging vertically from something rigid, and we'll hang a mass from the end of the spring. When we do, depending on the characteristics of the spring itself and the weight of the mass, the spring may stretch, and it may stretch and compress up and down, until eventually it reaches some motionless equilibrium point.



Hooke's Law tells us that the spring exerts force on the mass as it tries to return to its unstretched state, and that force is proportional to the extra length of the spring between its unstretched state and its state of equilibrium with the mass, in other words, $F = ks$, where k is the **spring constant**.

There are two scenarios here that we want to consider. First, we'll look at a spring and mass system that's vibrating free of external motion, and

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then we'll look at a system where the motion is forced by some external driver.

Free motion, undamped and damped

If the spring and mass system is operating in a vacuum, such that there are no other external forces exerting themselves on the system (like friction from the environment, or the decay of the spring over time), then we say that the motion of the spring and mass is **free undamped motion**, or simple harmonic motion.

Using Newton's Second Law, we can model this position of the mass over time in relation to equilibrium as

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

where x is the distance between the mass and its position in equilibrium, k is the spring constant, m is the mass, and t is time. It's also common to substitute $\omega^2 = k/m$ to rewrite this equation as

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

The reason is because, in this form, we can say that the general solution is

$$x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$$

Keep in mind, regardless of how we express the second order equation, x is positive when the mass is below its equilibrium position (the spring is

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stretched further than equilibrium) and negative when the mass is above its equilibrium position (the spring is less stretched than at equilibrium).

Let's do an example where we find the general solution to the second order equation.

Example

A mass weighing 10 pounds stretches a spring 4 ft. Find the equation that models the motion of the mass if we release the mass when $t = 0$ from a position 3 feet above equilibrium, with a downward velocity of 1 ft/s.

First we'll use Hooke's Law to find the spring constant k.

$$F = ks$$

$$10 = k(4)$$

$$k = 2.5 \text{ lb/ft}$$

To convert weight into mass, we'll use $m = W/g = 10/32 = 5/16$ slug. Plugging everything we have into the second order equation, we get

$$m\frac{d^2x}{dt^2} + kx = 0$$

$$\frac{5}{16}\left(\frac{d^2x}{dt^2}\right) + 2.5x = 0$$

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$$\frac{d^2x}{dt^2} + 8x = 0$$

From this equation, we see that $\omega = \sqrt{8} = 2\sqrt{2}$, which means that the general solution is given by

$$x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$$

$$x(t) = c_1 \cos(2\sqrt{2}t) + c_2 \sin(2\sqrt{2}t)$$

Its derivative is

$$x'(t) = -2\sqrt{2}c_1 \sin(2\sqrt{2}t) + 2\sqrt{2}c_2 \cos(2\sqrt{2}t)$$

The question states that the initial position is $x(0) = -3$ and the initial velocity is $x'(0) = 1$, so we'll plug these into $x(t)$ and $x'(t)$, and we get

$$-3 = c_1 \cos(2\sqrt{2}(0)) + c_2 \sin(2\sqrt{2}(0))$$

$$c_1 = -3$$

and

$$1 = -2\sqrt{2}c_1 \sin(2\sqrt{2}(0)) + 2\sqrt{2}c_2 \cos(2\sqrt{2}(0))$$

$$c_2 = \frac{\sqrt{2}}{4}$$

So the equation modeling the motion of this spring and mass system, with these particular initial conditions for velocity and position, is given by

$$x(t) = -3 \cos(2\sqrt{2}t) + \frac{\sqrt{2}}{4} \sin(2\sqrt{2}t)$$

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$$x(t) = \frac{\sqrt{2}}{4}\sin(2\sqrt{2}t) - 3\cos(2\sqrt{2}t)$$

Let's talk now about the scenario of **free damped motion**, in which we have free motion that's being damped, like by a liquid that surrounds the system, or by an actual damping device that's pushing back on the mass.

The second order differential equation we used to model free damped motion is

$$\frac{d^2x}{dt^2} + \frac{\beta}{m}\left(\frac{dx}{dt}\right) + \frac{k}{m}x = 0$$

or

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \omega^2 x = 0$$

where $2\lambda = \beta/m$ and $\omega^2 = k/m$. Again, just like the substitution $\omega^2 = k/m$ gave us a more convenient way to express the general solution, the substitution $2\lambda = \beta/m$ gives us a more convenient way to express the roots of the associated characteristic equation, $r^2 + 2\lambda r + \omega^2 = 0$.

$$r_{1,2} = -\lambda \pm \sqrt{\lambda^2 - \omega^2}$$

Given this pair of roots,

- if $\lambda^2 - \omega^2 > 0$, the system is **overdamped** and the general solution is

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$$x(t) = e^{-\lambda t}(c_1 e^{\sqrt{\lambda^2 - \omega^2}t} + c_2 e^{-\sqrt{\lambda^2 - \omega^2}t})$$

- if $\lambda^2 - \omega^2 = 0$, the system is **critically damped** and the general solution is

$$x(t) = c_1 e^{m_1 t} + c_2 t e^{m_1 t}$$

$$x(t) = e^{-\lambda t}(c_1 + c_2 t)$$

- if $\lambda^2 - \omega^2 < 0$, the system is **underdamped** and the general solution is

$$x(t) = e^{-\lambda t}(c_1 \cos\sqrt{\omega^2 - \lambda^2}t + c_2 \sin\sqrt{\omega^2 - \lambda^2}t)$$

Let's do an example where we find the solution for a free damped system.

Example

A mass weighing 4 pounds stretches a spring 2 feet, and a damping force that's triple the instantaneous velocity is acting on the spring/mass system. Find an equation that models the motion of the mass if it's initially released from $1/2$ foot below equilibrium with an upward velocity of 1 ft/s.

Hooke's Law tells us that the spring constant is

$$F = ks$$

$$4 = k(2)$$

$$k = 2 \text{ lb/ft}$$

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Now we'll use $W = mg$ to convert the weight into mass.

$$W = mg$$

$$4 = m(32)$$

$$m = \frac{1}{8} \text{ slug}$$

With the spring constant and the mass, and $\beta = 3$ since the damping force is triple the instantaneous velocity, we can plug everything into the differential equation.

$$\frac{d^2x}{dt^2} + \frac{\beta}{m}\left(\frac{dx}{dt}\right) + \frac{k}{m}x = 0$$

$$\frac{d^2x}{dt^2} + \frac{3}{\frac{1}{8}}\left(\frac{dx}{dt}\right) + \frac{2}{\frac{1}{8}}x = 0$$

$$\frac{d^2x}{dt^2} + 24\left(\frac{dx}{dt}\right) + 16x = 0$$

The associated characteristic equation and its roots are

$$r^2 + 24r + 16 = 0$$

$$r = -12 \pm 8\sqrt{2}$$

Because we find roots in which $\lambda^2 - \omega^2 > 0$, the spring and mass system is overdamped and the general solution is

$$x(t) = e^{-\lambda t}(c_1 e^{\sqrt{\lambda^2 - \omega^2}t} + c_2 e^{-\sqrt{\lambda^2 - \omega^2}t})$$

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$$x(t) = e^{-12t}(c_1 e^{\sqrt{12^2 - 16}t} + c_2 e^{-\sqrt{12^2 - 16}t})$$

$$x(t) = e^{-12t}(c_1 e^{\sqrt{128}t} + c_2 e^{-\sqrt{128}t})$$

$$x(t) = e^{-12t}(c_1 e^{8\sqrt{2}t} + c_2 e^{-8\sqrt{2}t})$$

Driven motion

When we add some driver of motion into the system, we often call it the forcing function $f(t)$, and we rewrite Newton's second law as

$$\frac{d^2x}{dt^2} + \frac{\beta}{m}\left(\frac{dx}{dt}\right) + \frac{k}{m}x = \frac{f(t)}{m}$$

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \omega^2 x = F(t)$$

Since this is now a nonhomogeneous second order equation, we'd just need to use the method of undetermined coefficients or the method of variation of parameters to find the solution.

For instance, using undetermined coefficients, we'd follow the same steps we're used to with that method. We'd start by finding the complementary solution as the solution to the associated homogeneous equation. Then we'd make a guess for the particular solution and apply undetermined coefficients. To get the general solution, we'd sum the complementary and particular solutions.

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And if we have initial conditions, we can apply them to the general solution and its derivative in order to solve for the coefficients c_1 and c_2.

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